Report_analisi_esplorativa

2026-04-14

The cohort is composed by 30 patients affected by multiple sclerosis recruited from multiple centers (Clerico, Molinette, Napoli, CRESM). All of them were treated with cladribine (Mavenclad). The longitudinal clinical data were collected at 6-month time-points for 3 years.

The population is predominantly female, with 20 females (66.7%) and 9 males (30.0%). At baseline (T0), the mean age was 37.37 years (± 9.89), with a median of 38.50 years, ranging from 20 to 55 years, indicating a relatively young cohort. Disease activity was predominantly classified as higher activity. Disability at baseline, measured by EDSS, showed generally low values reflected by a median of 1.5 and 26.7% of patients with a EDSS 1. Regarding treatment history, 18 patients (60.0%) were naive, while 11 (36.7%) had received prior therapies. Glucocorticoid treatment before T0 was reported in 12 patients (40.0%), whereas 6 patients (20.0%) had not received such treatment and data were missing for 12 subjects.

Regarding lifestyle factors, the mean body mass index (BMI) was 23.54 ± 4.85, ranging from 16.80 to 32.10, with 9 missing values. Adherence to the Mediterranean diet, assessed by the MEDAS score (from 0 to 100), showed a mean value of 50.55 ± 15.18, with a wide range (35.00–85.70) and 8 missing observations. 7 patients (23.3%) are smokers and 17 (56.7%) are non-smokers, with 6 missing values. Physical activity is heterogeneously distributed among the cohort with 5 (16.7%) patients classified as active, 7 (23.3%) as partially active and 9 (30%) as sedentary, with 9 missing values.

```
####tabella riassuntiva stile paper

#variabili numeriche
tab_num <- merged_data %>%
  dplyr::select(where(is.numeric)) %>%
  dplyr::select(-any_of(c("percent_th17", "treg_cd39_pos", "treg_il10_pos"))) %>%
  tidyr::pivot_longer(everything(), names_to = "variable", values_to = "val") %>%
  mutate(variable = factor(variable, levels = unique(names(merged_data)))) %>%
  dplyr::group_by(variable) %>%
  dplyr::summarise(
    All = paste0(round(mean(val, na.rm = TRUE), 2), " ± ", round(sd(val, na.rm = TRUE), 2)),
    `N missing` = as.character(sum(is.na(val))),
    .groups = "drop"
  )

#variabili categoriche
vars_cat <- merged_data %>%
  dplyr::select(where(~is.factor(.) | is.character(.) | is.ordered(.)),
                -any_of(c("ID", "NOTE", "nuova_terapia", "previous_therapy", "sample_type"))) %>%
  colnames()
lista_cat <- lapply(vars_cat, function(v) {
  miss_count <- as.character(sum(is.na(merged_data[[v]])))

levels_df <- merged_data %>%
    dplyr::group_by(level = as.character(!!sym(v))) %>%
    dplyr::summarise(n = n(), .groups = "drop") %>%
    dplyr::filter(!is.na(level)) %>%
```

```
    dplyr::mutate(
      All = paste0(n, " (", round((n / nrow(merged_data)) * 100, 1), "%)"),
      `N missing` = ""
    ) %>%
    dplyr::mutate(variable = paste0("   ", level)) %>%
    dplyr::select(variable, All, `N missing`)

  bind_rows(
    tibble(variable = v, All = "", `N missing` = miss_count),
    levels_df
  )
})

tab_cat <- bind_rows(lista_cat)

#unione
summary_table <- bind_rows(tab_num, tab_cat)

# Prima di kable, puliamo TUTTA la tabella da caratteri che rompono LaTeX
summary_table$variable <- gsub("_", " ", summary_table$variable)
summary_table$All <- gsub("%", "\\\\%", summary_table$All) # Protegge il simbolo %

knitr::kable(summary_table, caption = "Cohort Summary Table")
```

Table 1: Cohort Summary Table

variable	All	N missing
MEDAS	50.55 ± 15.18	8
cortisone T6	1.29 ± 3.72	13
cortisone T12	0 ± 0	10
cortisone T18	0 ± 0	14
cortisone T24	0.54 ± 1.33	17
cortisone T30	0.58 ± 1.38	18
cortisone T36	0.6 ± 1.34	25
age at t0	37.37 ± 9.89	0
bmi	23.54 ± 4.85	9
dna quantification	105.36 ± 60.25	28
days gc collection	187.09 ± 237.89	19
age	41.53 ± 9.87	0
CENTRO		0
CLERICO	22 (73.3%)	
CRESM	4 (13.3%)	
MOLINETTE	1 (3.3%)	
NAPOLI	3 (10%)	
EDSS T0		1
0	2 (6.7%)	
1	8 (26.7%)	
1.5	4 (13.3%)	
2	6 (20%)	
2.5	4 (13.3%)	
3.5	2 (6.7%)	
4	1 (3.3%)	
5	2 (6.7%)	

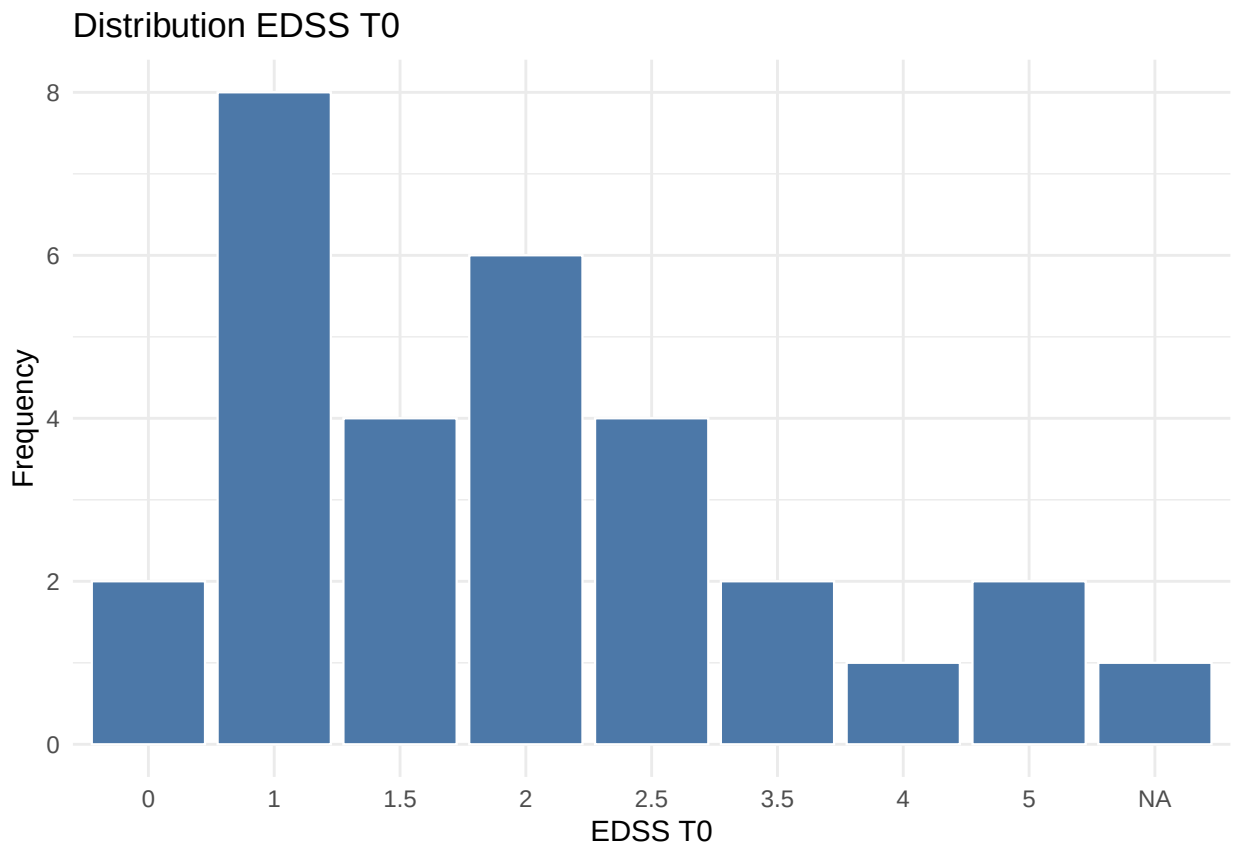
variable	All	N missing
EDSS T6		15
0	2 (6.7%)	
1	5 (16.7%)	
1.5	2 (6.7%)	
2	3 (10%)	
2.5	1 (3.3%)	
3	1 (3.3%)	
4.5	1 (3.3%)	
lesioni T6		13
No	15 (50%)	
Yes	2 (6.7%)	
relapse T6		13
No	14 (46.7%)	
Yes	3 (10%)	
EDSS T12		2
0	5 (16.7%)	
1	8 (26.7%)	
1.5	4 (13.3%)	
2	3 (10%)	
2.5	2 (6.7%)	
3	1 (3.3%)	
3.5	1 (3.3%)	
4	1 (3.3%)	
4.5	2 (6.7%)	
6.5	1 (3.3%)	
lesioni T12		6
No	21 (70%)	
Yes	3 (10%)	
relapse T12		6
No	24 (80%)	
EDSS T18		16
0	2 (6.7%)	
1	4 (13.3%)	
1.5	3 (10%)	
2	2 (6.7%)	
4	1 (3.3%)	
4.5	1 (3.3%)	
6.5	1 (3.3%)	
lesioni T18		14
No	13 (43.3%)	
Yes	3 (10%)	
relapse T18		14
No	15 (50%)	
Yes	1 (3.3%)	
EDSS T24		12
0	3 (10%)	
1	4 (13.3%)	
1.5	3 (10%)	
2	3 (10%)	
2.5	2 (6.7%)	
4	2 (6.7%)	
6.5	1 (3.3%)	

variable	All	N missing
lesioni T24		13
No	14 (46.7%)	
Yes	3 (10%)	
relapse T24		13
No	15 (50%)	
Yes	2 (6.7%)	
EDSS T30		18
0	3 (10%)	
1	2 (6.7%)	
1.5	3 (10%)	
2	2 (6.7%)	
4.5	1 (3.3%)	
6.5	1 (3.3%)	
lesioni T30		18
No	9 (30%)	
Yes	3 (10%)	
relapse T30		18
No	9 (30%)	
Yes	3 (10%)	
EDSS T36		25
0	1 (3.3%)	
1	2 (6.7%)	
1.5	1 (3.3%)	
6.5	1 (3.3%)	
lesioni T36		25
No	4 (13.3%)	
Yes	1 (3.3%)	
relapse T36		25
No	5 (16.7%)	
cambio terapia		9
No	14 (46.7%)	
Yes	7 (23.3%)	
therapy		0
cladribine	30 (100%)	
naive		1
No	11 (36.7%)	
Yes	18 (60%)	
sex		1
F	20 (66.7%)	
M	9 (30%)	
smoking habit		6
No	17 (56.7%)	
Yes	7 (23.3%)	
physical activity		9
active	5 (16.7%)	
partially active	7 (23.3%)	
sedentary	9 (30%)	
antibiotic use		12
No	16 (53.3%)	
Yes	2 (6.7%)	
sequencing batch		12
1	16 (53.3%)	

variable	All	N missing
3	2 (6.7%)	
spike in		12
No	16 (53.3%)	
Yes	2 (6.7%)	
glucocorticoid treatment		12
Positive	18 (60%)	
simplified batch		12
1-3	18 (60%)	
disease activity		12
higher activity	18 (60%)	
gc treatment		12
Negative	6 (20%)	
Positive	12 (40%)	
lesion burden		12
No	5 (16.7%)	
Yes	13 (43.3%)	
bone marrow lesions		12
No	4 (13.3%)	
Yes	14 (46.7%)	
subtentorial lesions		13
No	11 (36.7%)	
Yes	6 (20%)	
gadolinium contrast		12
No	14 (46.7%)	
Yes	4 (13.3%)	
clinical presentation		12
bone marrow	4 (13.3%)	
brain stem	1 (3.3%)	
motor impairment	1 (3.3%)	
multifocal	1 (3.3%)	
norb/sensoriale	1 (3.3%)	
optic neuritis	3 (10%)	
sensoriale	2 (6.7%)	
sensory symptoms	4 (13.3%)	
tronco	1 (3.3%)	
full recovery		12
No	7 (23.3%)	
Yes	11 (36.7%)	
prognosis at onset		15
Negative	8 (26.7%)	
Positive	7 (23.3%)	

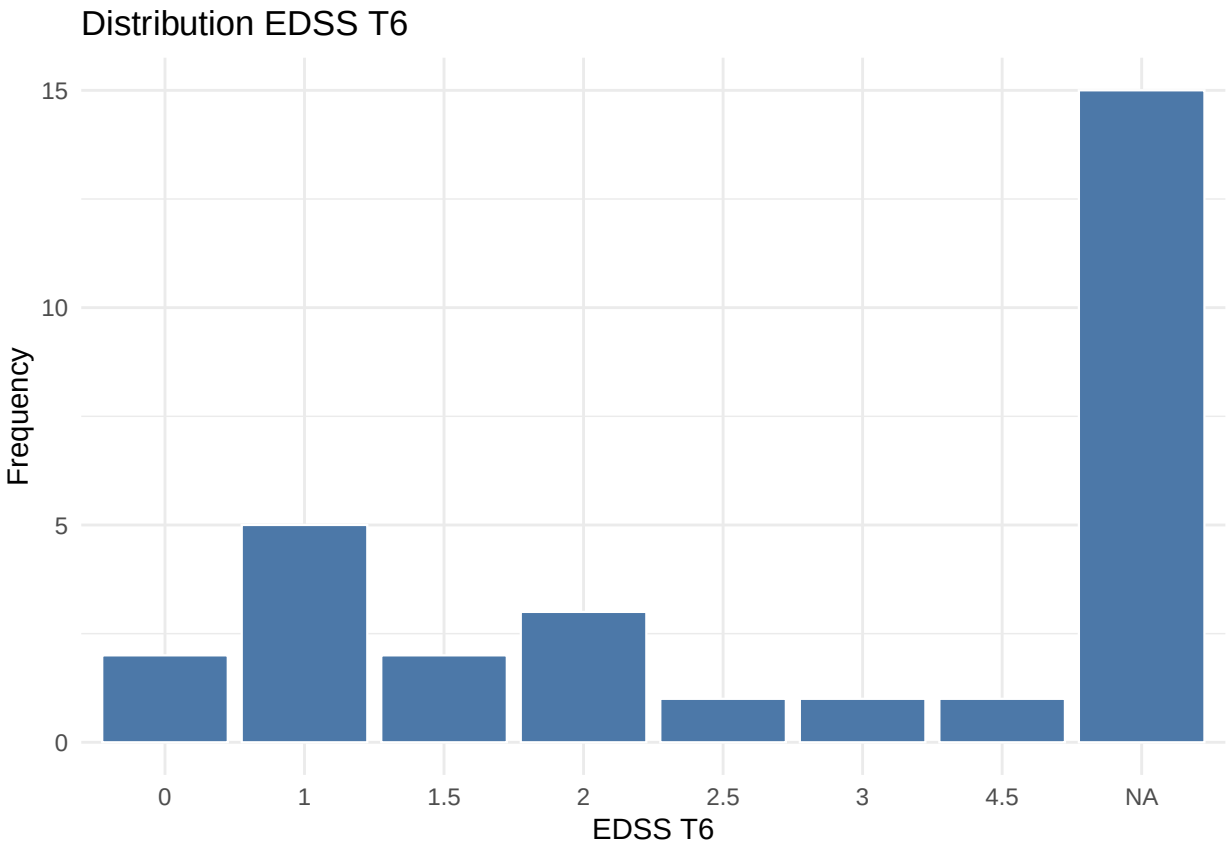
Summary graph of the distribution of the EDSS variable at time-point T0 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T0)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) + t
```



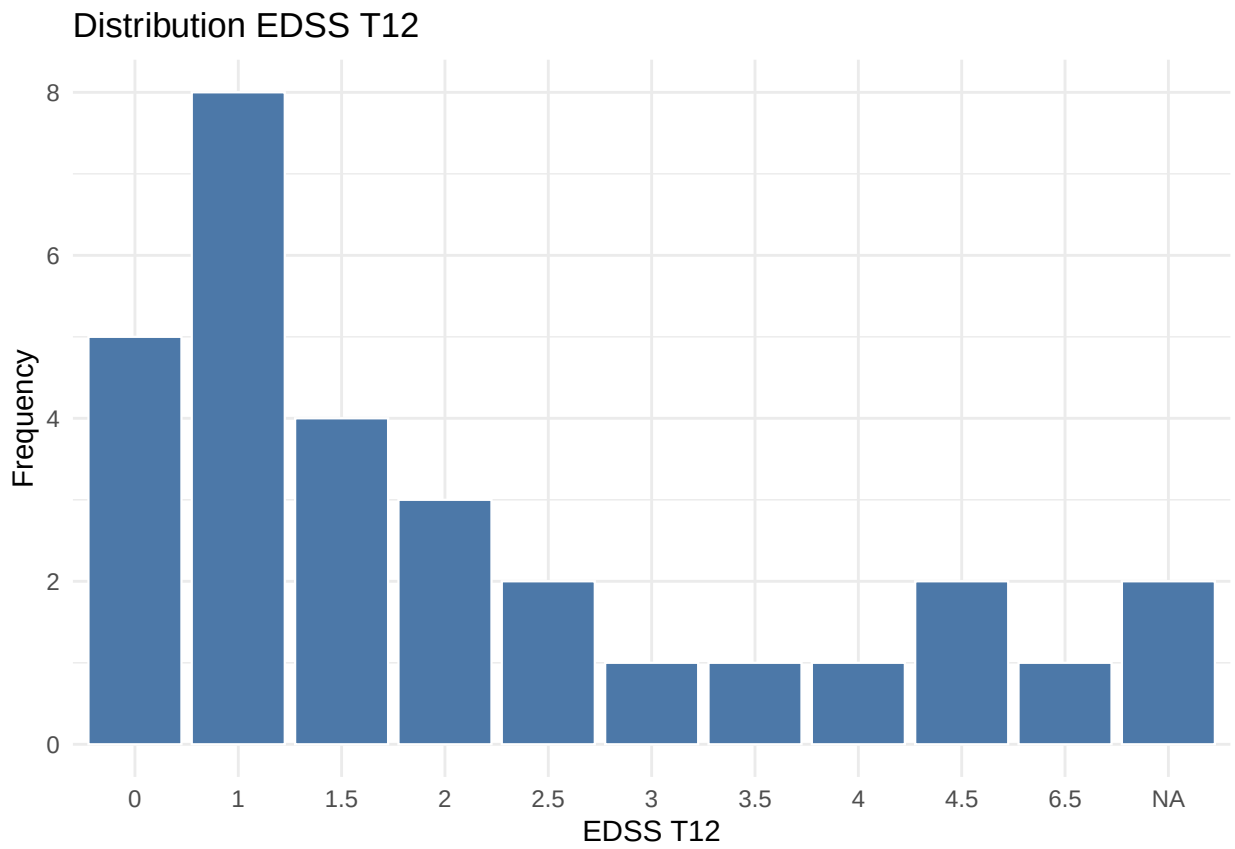
Summary graph of the distribution of the EDSS variable at time-point T6 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T6)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) + t
```



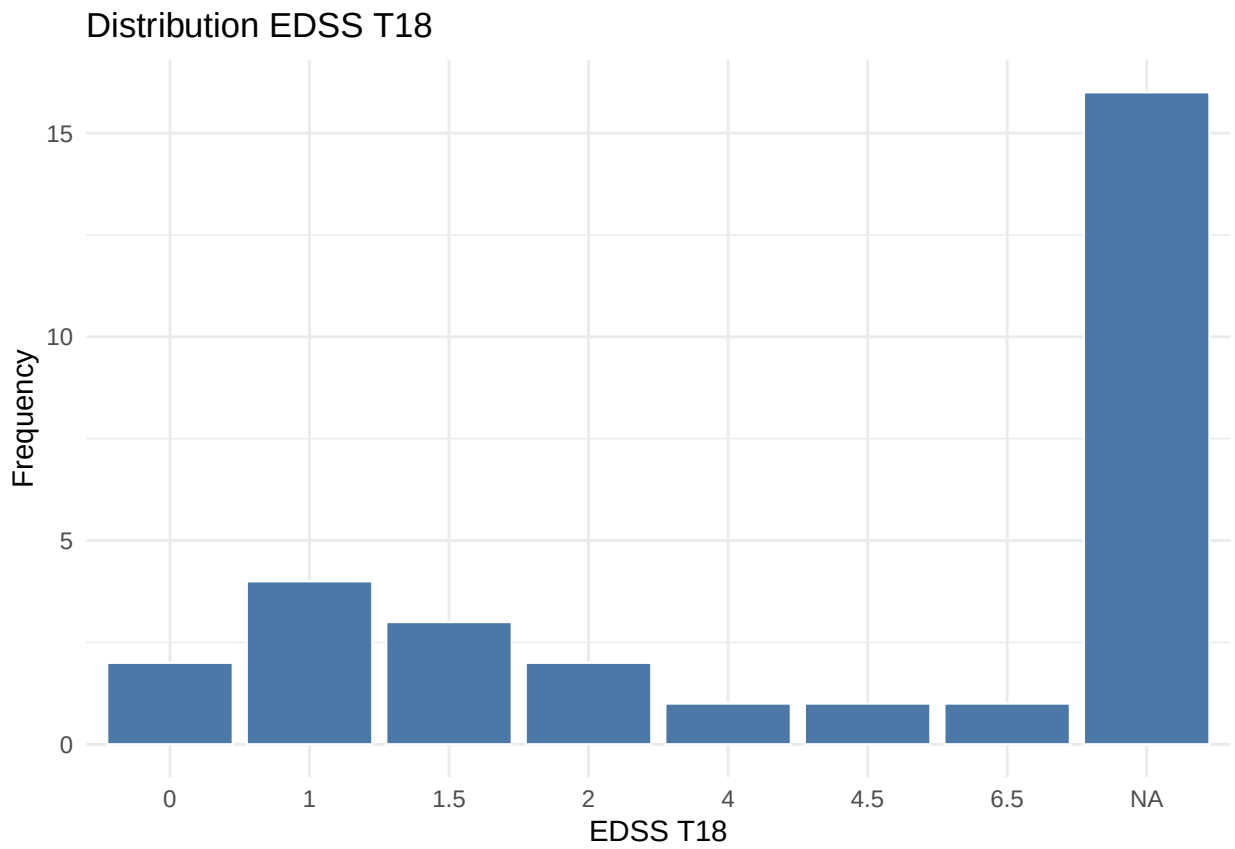
Summary graph of the distribution of the EDSS variable at time-point T12 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T12)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) +
```



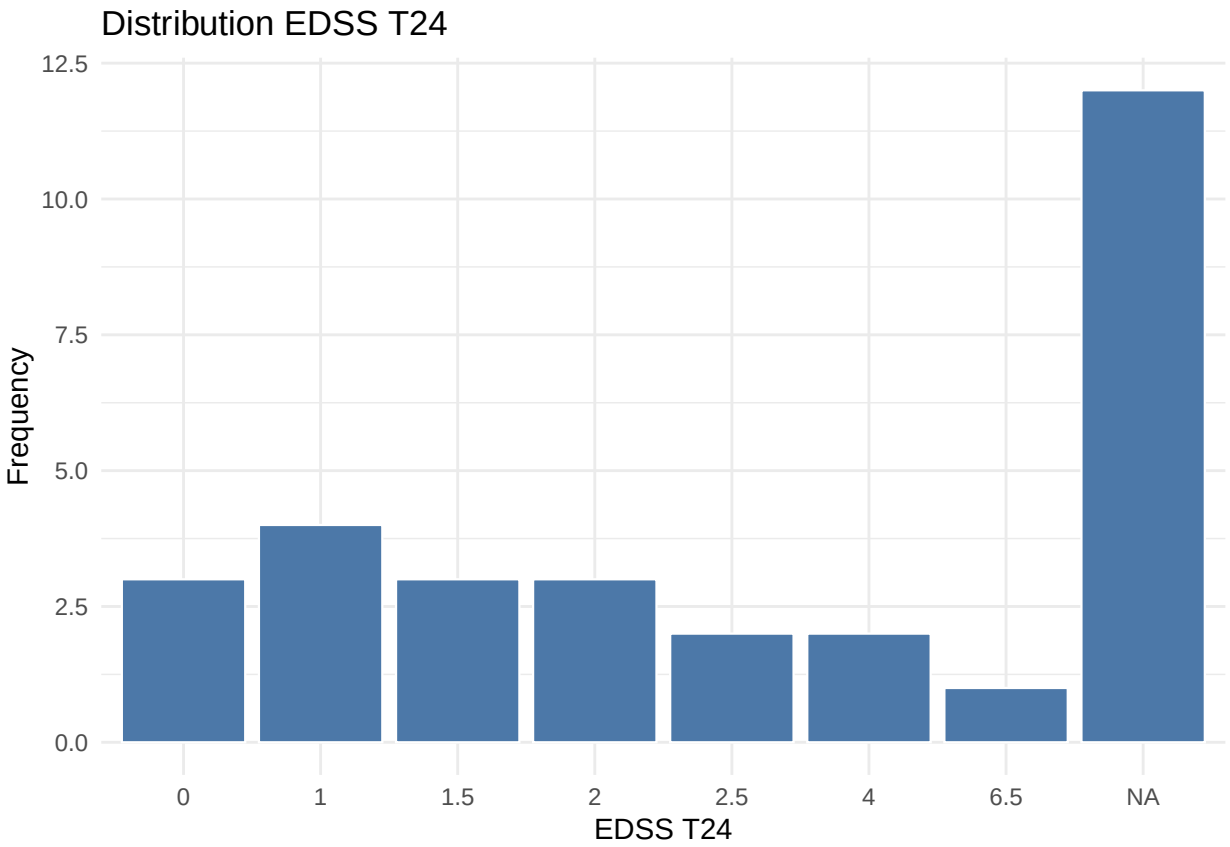
Summary graph of the distribution of the EDSS variable at time-point T18 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T18)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) +
```



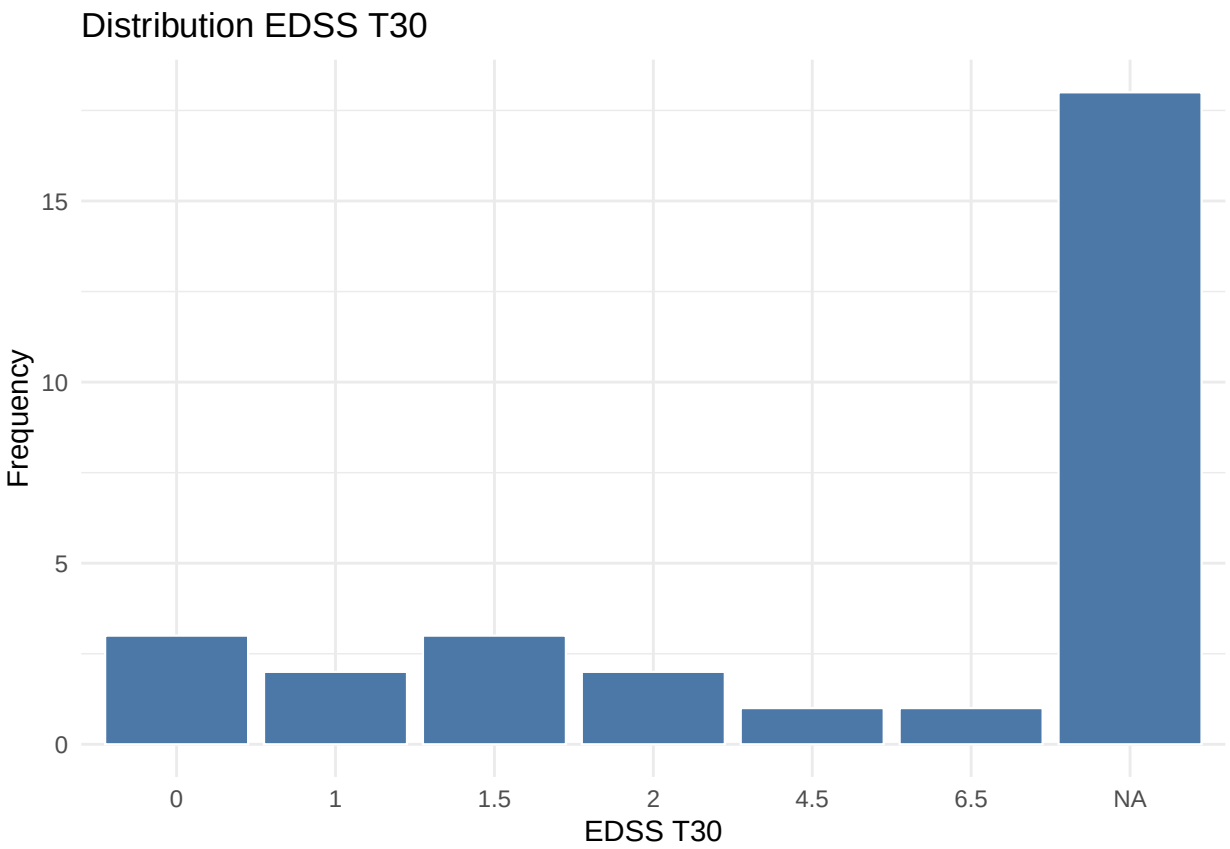
Summary graph of the distribution of the EDSS variable at time-point T24 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T24)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) +
```



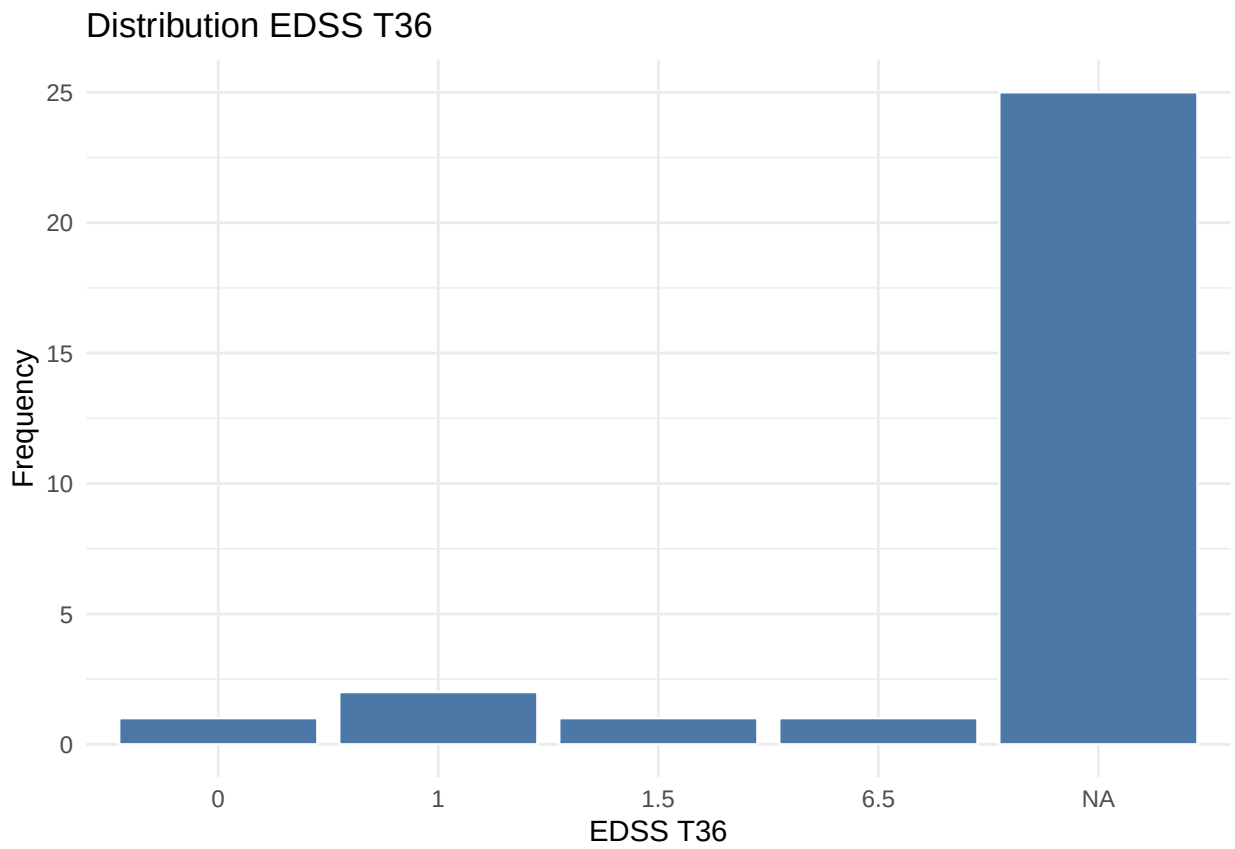
Summary graph of the distribution of the EDSS variable at time-point T30 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T30)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) +
```



Summary graph of the distribution of the EDSS variable at time-point T36 in the cohort:

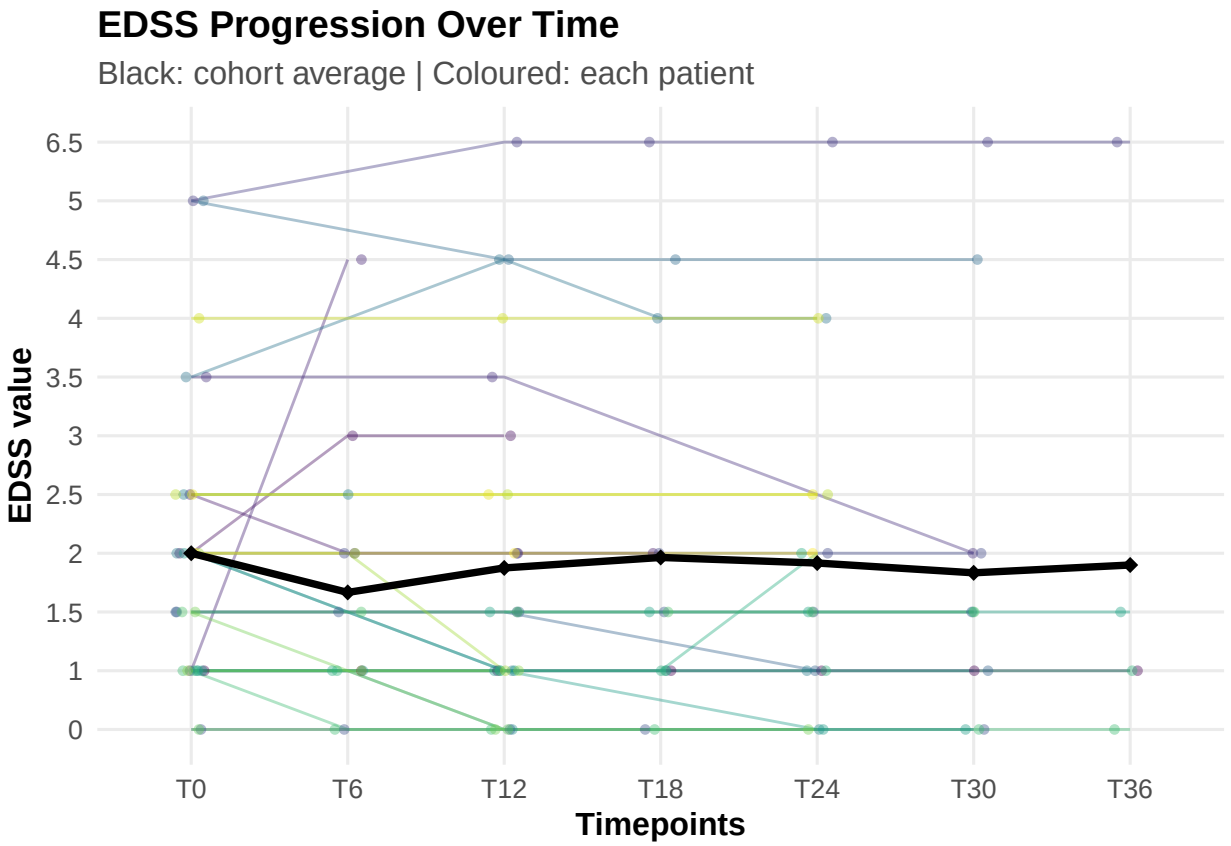
```
ggplot(merged_data, aes(x = EDSS_T36)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) +
```



Summary linepoint of the trend of the EDSS variable over time for each patient in the cohort:

```
#Preparazione dati
edss_long <- merged_data %>%
  select(ID, starts_with("EDSS_T")) %>%
  pivot_longer(cols = starts_with("EDSS_T"), names_to = "timepoint", values_to = "EDSS") %>%
  filter(!is.na(EDSS)) %>%
  mutate(timepoint = factor(timepoint,
                            levels = paste0("EDSS_T", c(0, 6, 12, 18, 24, 30, 36)),
                            labels = paste0("T", c(0, 6, 12, 18, 24, 30, 36))))

#linepoint
ggplot(edss_long, aes(x = timepoint, y = EDSS, group = ID)) +
  #Linee pazienti
  geom_line(aes(color = ID), alpha = 0.4, size = 0.5) +
  #jitter per più pazienti sullo stesso valore
  geom_point(aes(color = ID), alpha = 0.4, size = 1.2, position = position_jitter(width = 0.1, height =
  #Linea media
  stat_summary(aes(group = 1), fun = mean, geom = "line", size = 1.3, color = "black") +
  stat_summary(aes(group = 1), fun = mean, geom = "point", size = 2.5, color = "black", shape = 18) + s
  labs(title = "EDSS Progression Over Time", subtitle = "Black: cohort average | Coloured: each patient
  theme(panel.grid.minor = element_blank(), axis.title = element_text(face = "bold"), plot.title = eleme
```



Exploratory analysis of some relevant demographic variables: 1. EDSS and age at T0, positive and significant difference between groups only in T24 and T30 time-points (we have to consider that these two time-points have a larger number of NA).

```
# EDSS vs age_at_t0
cor.test(merged_data$age_at_t0, as.numeric(merged_data$EDSS_T0), method = "spearman", exact = FALSE) #
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$age_at_t0 and as.numeric(merged_data$EDSS_T0)
## S = 2958.8, p-value = 0.1547
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.2712387
```

```
cor.test(merged_data$age_at_t0, as.numeric(merged_data$EDSS_T12), method = "spearman", exact = FALSE) #
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$age_at_t0 and as.numeric(merged_data$EDSS_T12)
## S = 2579, p-value = 0.1286
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.2942045
```

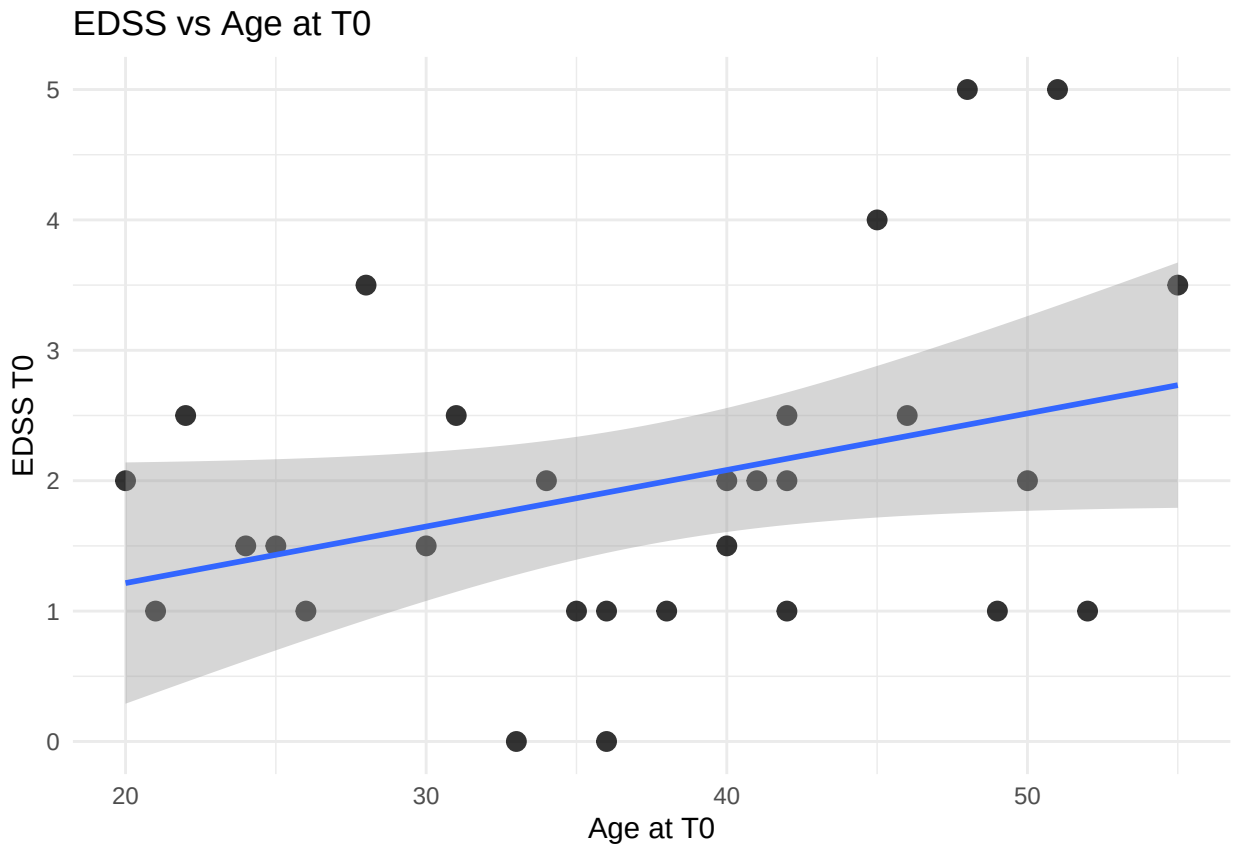
```r
cor.test(merged_data$age_at_t0, as.numeric(merged_data$EDSS_T24), method = "spearman", exact = FALSE) #
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$age_at_t0 and as.numeric(merged_data$EDSS_T24)
## S = 438.62, p-value = 0.01872
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.5473475
```

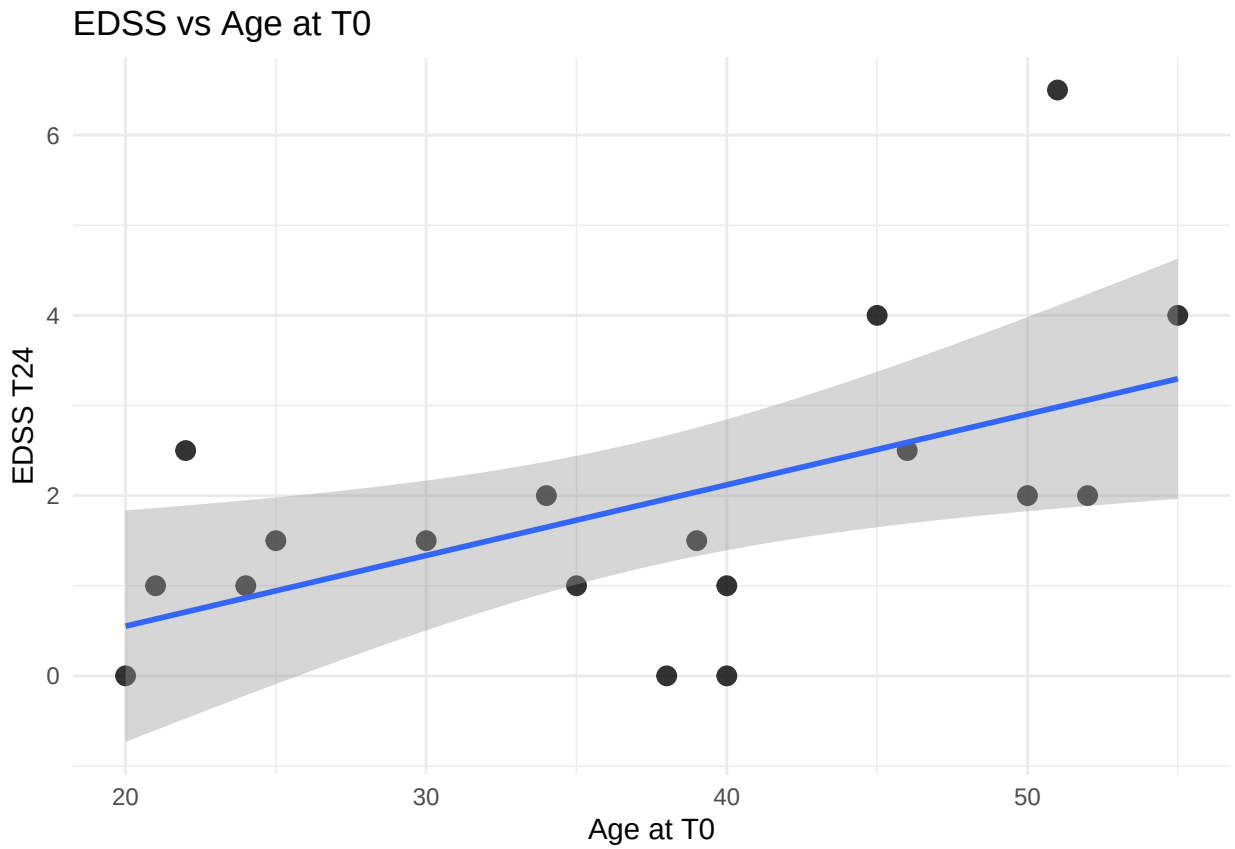
```r
cor.test(merged_data$age_at_t0, as.numeric(merged_data$EDSS_T30), method = "spearman", exact = FALSE) #
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$age_at_t0 and as.numeric(merged_data$EDSS_T30)
## S = 120.07, p-value = 0.04798
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.580163
```

```r
# linee di regressione EDSS vs age_at_t0
ggplot(merged_data, aes(x = age_at_t0, y = as.numeric(as.character(EDSS_T0)))) + geom_point(size = 3, a
```



```r
ggplot(merged_data, aes(x = age_at_t0, y = as.numeric(as.character(EDSS_T24)))) + geom_point(size = 3, a
```



2. EDSS and sex, non-significant difference between groups at all time points.

```r
# EDSS vs sex ##correlazione non significativa in nessun timepoint
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ sex, data = merged_data, exact = FALSE) # differenza no
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by sex
## W = 88, p-value = 0.6979
## alternative hypothesis: true location shift is not equal to 0
```

```r
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ sex, data = merged_data, exact = FALSE) # differenza n
```

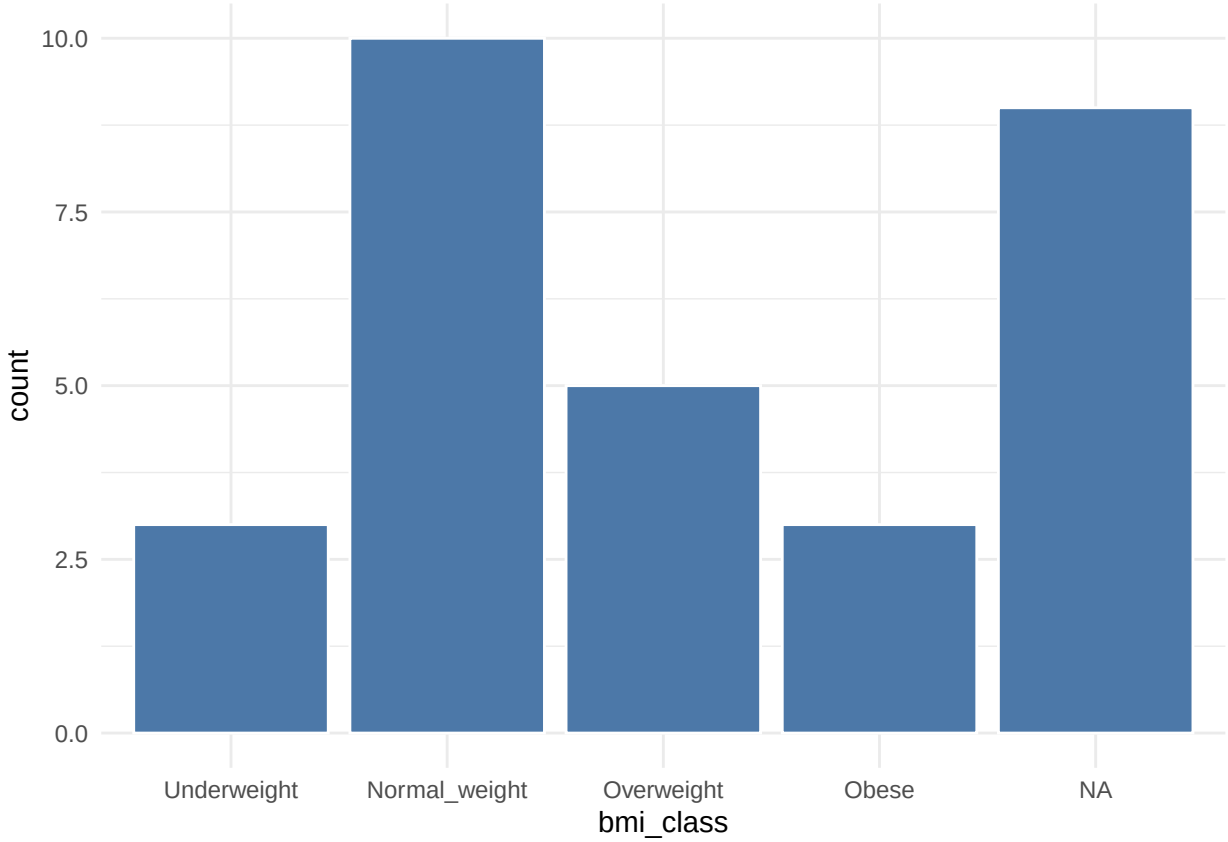
```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by sex
## W = 84, p-value = 0.6851
## alternative hypothesis: true location shift is not equal to 0
```

Exploratory analysis of some relevant lifestyle factors: 1. Summary graph of the distribution of the BMI variable in the cohort:

```r
merged_data$bmi_class <- factor(case_when(
    merged_data$bmi < 18.5 ~ "Underweight",
```

```
    merged_data$bmi >= 18.5 & merged_data$bmi < 25 ~ "Normal_weight",
    merged_data$bmi >= 25 & merged_data$bmi < 30 ~ "Overweight",
    merged_data$bmi >= 30 ~ "Obese",
    TRUE ~ NA_character_),
  levels = c("Underweight", "Normal_weight", "Overweight", "Obese"),ordered = TRUE)

ggplot(merged_data, aes(x = bmi_class)) + geom_bar(fill = "#4C78A8", color = "white") + theme_minimal()
```



1.1. EDSS and BMI, non-significant difference between groups at all time points.

```
# EDSS vs bmi
cor.test(merged_data$bmi, as.numeric(merged_data$EDSS_T0), method = "spearman", exact = FALSE)
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$bmi and as.numeric(merged_data$EDSS_T0)
## S = 972.79, p-value = 0.1004
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.3683168
```

```
cor.test(merged_data$bmi, as.numeric(merged_data$EDSS_T12), method = "spearman", exact = FALSE)
```

```
##
##  Spearman's rank correlation rho
```

```
##
## data:  merged_data$bmi and as.numeric(merged_data$EDSS_T12)
## S = 932.13, p-value = 0.2001
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.2991508
```

2. EDSS and smoking habit, non-significant difference between groups at all time points.

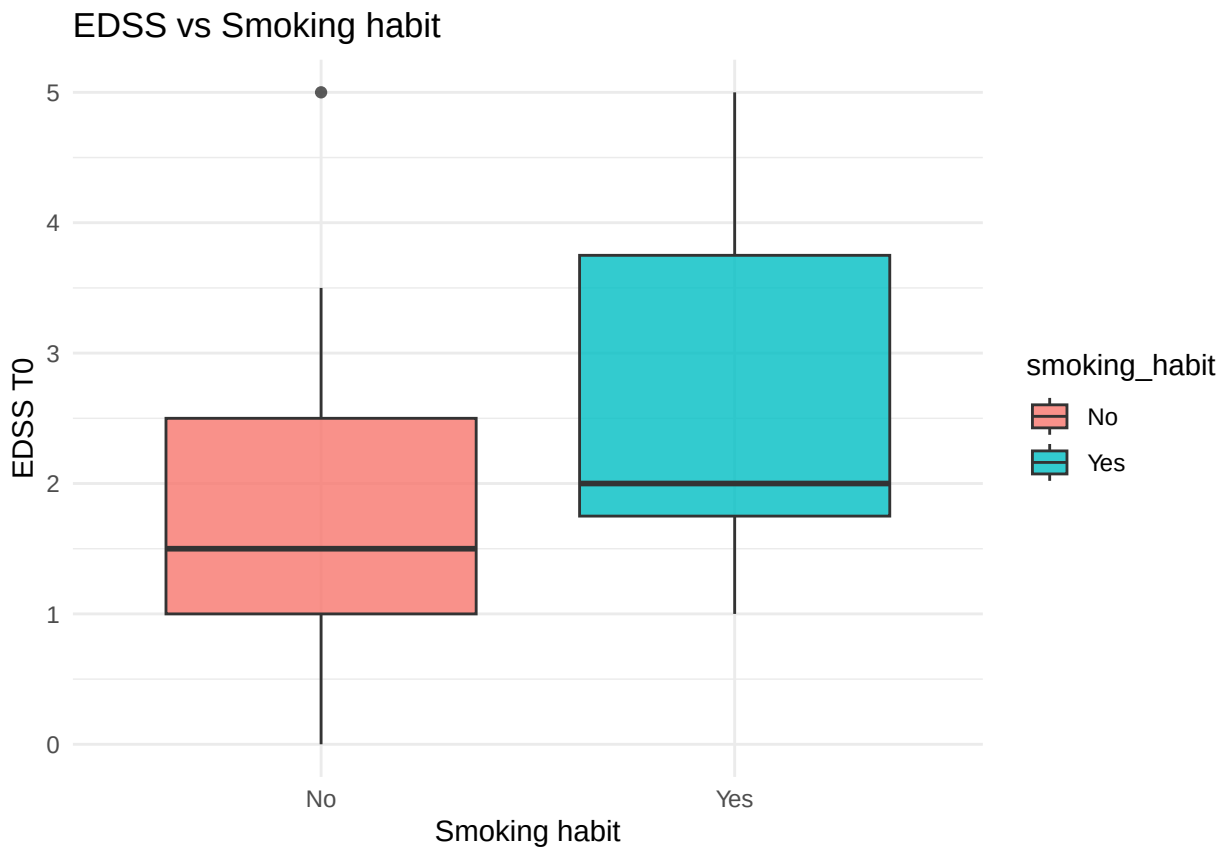
```
# EDSS vs smoking_habit #correlazione non significativa in nessun timepoint
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ smoking_habit, data = merged_data, exact = FALSE) # dif
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by smoking_habit
## W = 38.5, p-value = 0.1874
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ smoking_habit, data = merged_data, exact = FALSE) # di
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by smoking_habit
## W = 38.5, p-value = 0.2488
## alternative hypothesis: true location shift is not equal to 0
```

```
ggplot(subset(merged_data, !is.na(smoking_habit) & !is.na(EDSS_T0)), aes(x = smoking_habit, y = as.nume
```



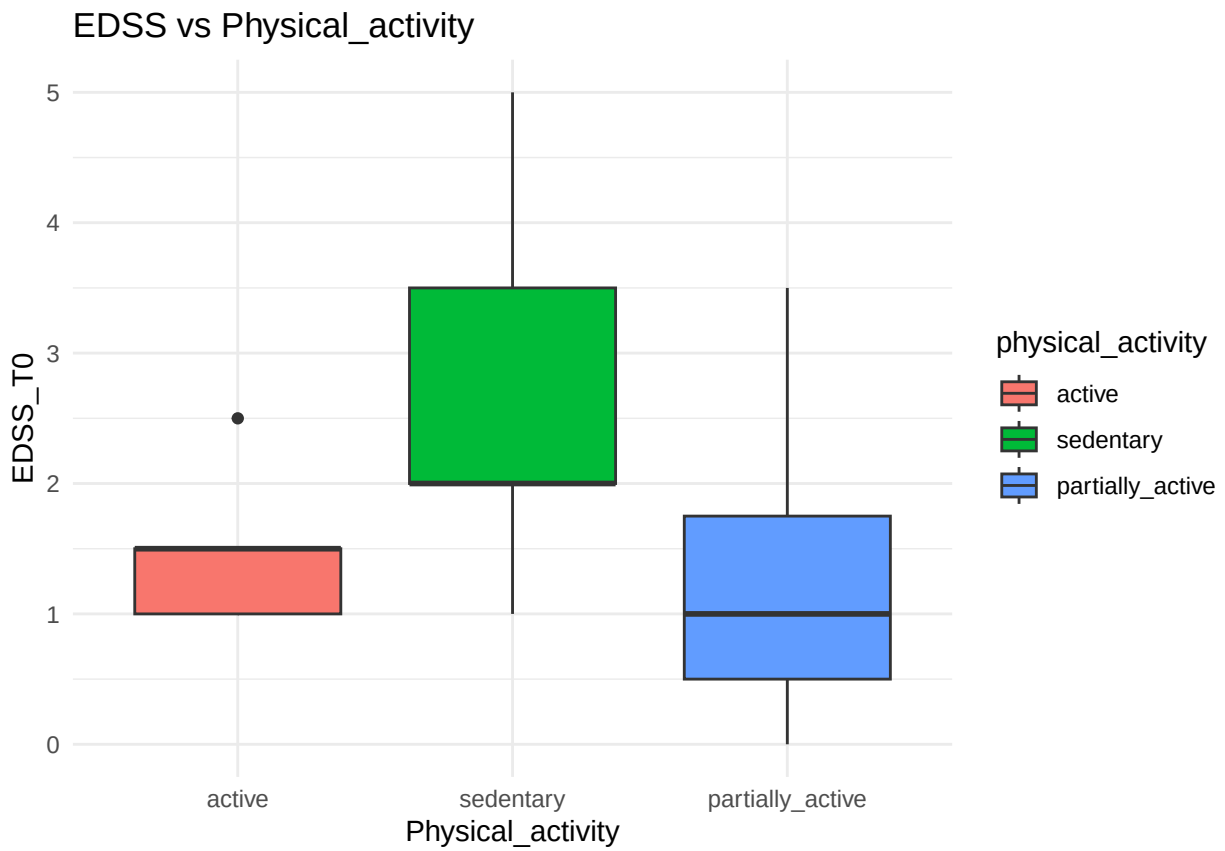
3. EDSS and physical activity, a possible trend can be seen in the graph between the sedentary lifestyle and an increase in the disability index (EDSS), but the statistical test is not significant.

```
## EDSS vs physical_activity

#T0
kruskal.test(as.numeric(as.character(EDSS_T0)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T0)) by physical_activity
## Kruskal-Wallis chi-squared = 5.168, df = 2, p-value = 0.07547
```

```
ggplot(subset(merged_data, !is.na(physical_activity) & !is.na(EDSS_T0)), aes(x = physical_activity, y =
```




```
kruskal.test(as.numeric(as.character(EDSS_T6)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T6)) by physical_activity
## Kruskal-Wallis chi-squared = 2.1859, df = 2, p-value = 0.3352
```

```
kruskal.test(as.numeric(as.character(EDSS_T12)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T12)) by physical_activity
## Kruskal-Wallis chi-squared = 4.1219, df = 2, p-value = 0.1273
```

```
kruskal.test(as.numeric(as.character(EDSS_T18)) ~ physical_activity, data = merged_data)
```

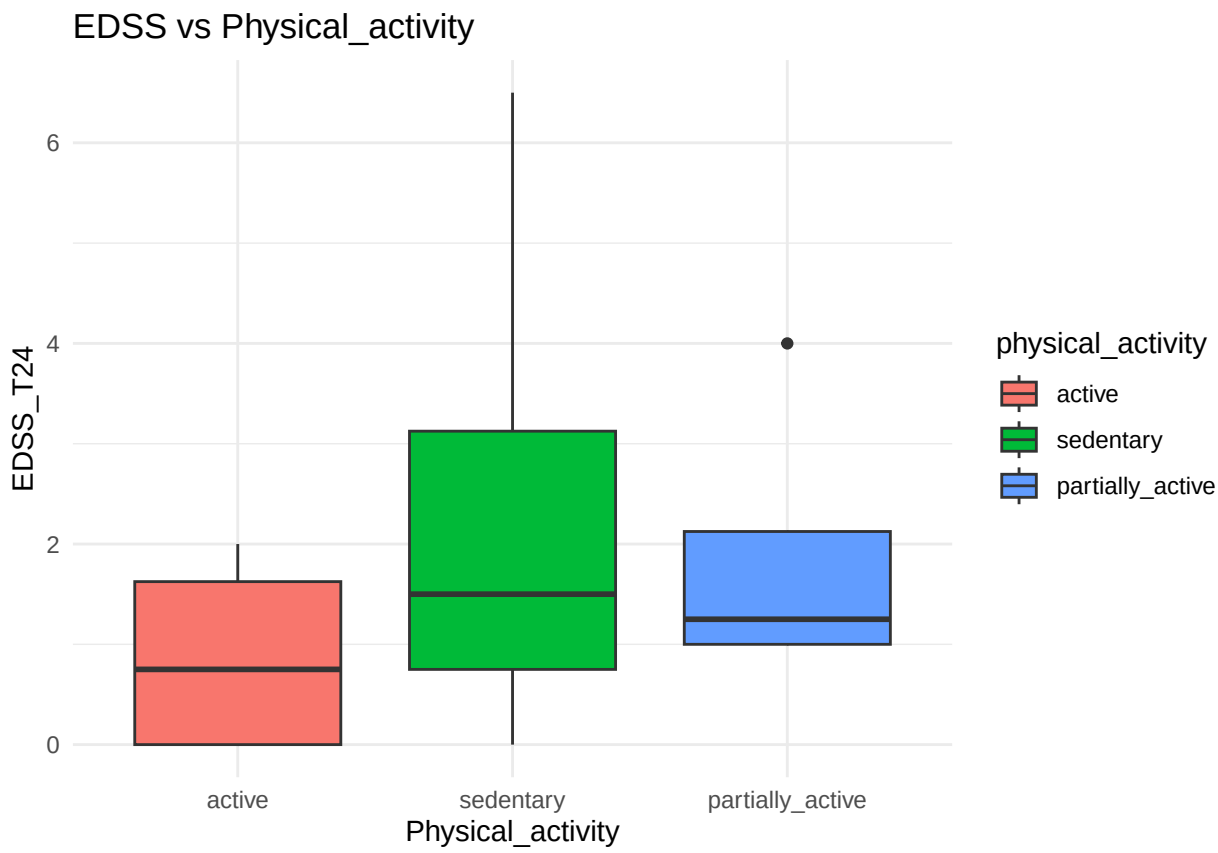
```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T18)) by physical_activity
## Kruskal-Wallis chi-squared = 1.5596, df = 2, p-value = 0.4585
```

```
#T24
kruskal.test(as.numeric(as.character(EDSS_T24)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
```

```
##
## data:  as.numeric(as.character(EDSS_T24)) by physical_activity
## Kruskal-Wallis chi-squared = 0.74728, df = 2, p-value = 0.6882
```

```
ggplot(subset(merged_data, !is.na(physical_activity) & !is.na(EDSS_T24)), aes(x = physical_activity, y =
```




```
kruskal.test(as.numeric(as.character(EDSS_T30)) ~ physical_activity, data = merged_data)
```

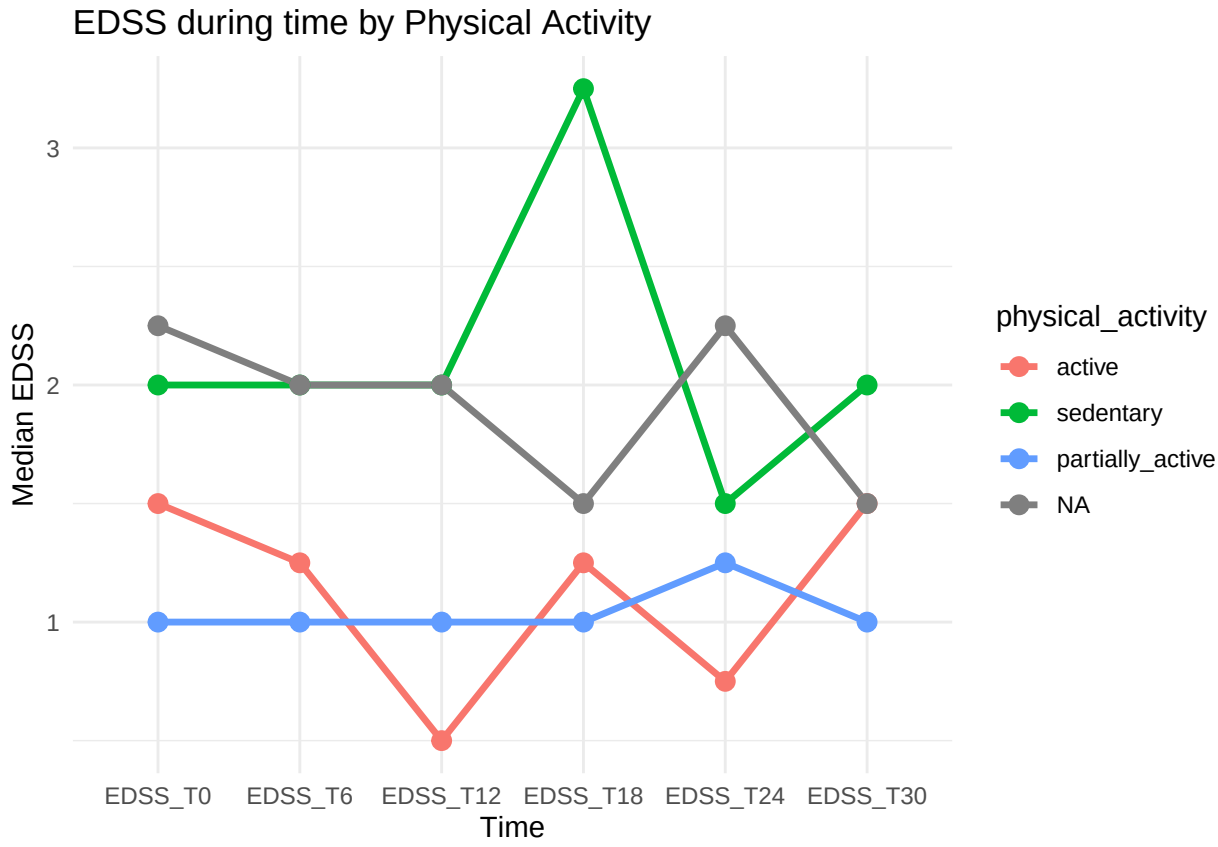
```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T30)) by physical_activity
## Kruskal-Wallis chi-squared = 1.0776, df = 2, p-value = 0.5835
```

```
kruskal.test(as.numeric(as.character(EDSS_T36)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T36)) by physical_activity
## Kruskal-Wallis chi-squared = 0, df = 1, p-value = 1
```

```
#andamento totale
edss_long <- merged_data %>%
  select(ID, smoking_habit, sex, naive, physical_activity, EDSS_T0, EDSS_T6, EDSS_T12, EDSS_T18, EDSS_T
  pivot_longer(cols = starts_with("EDSS"), names_to = "time", values_to = "EDSS_val") %>%
  mutate(EDSS_val = as.numeric(as.character(EDSS_val)), sex = trimws(as.character(sex)), time = factor(
  filter(!is.na(EDSS_val))
```

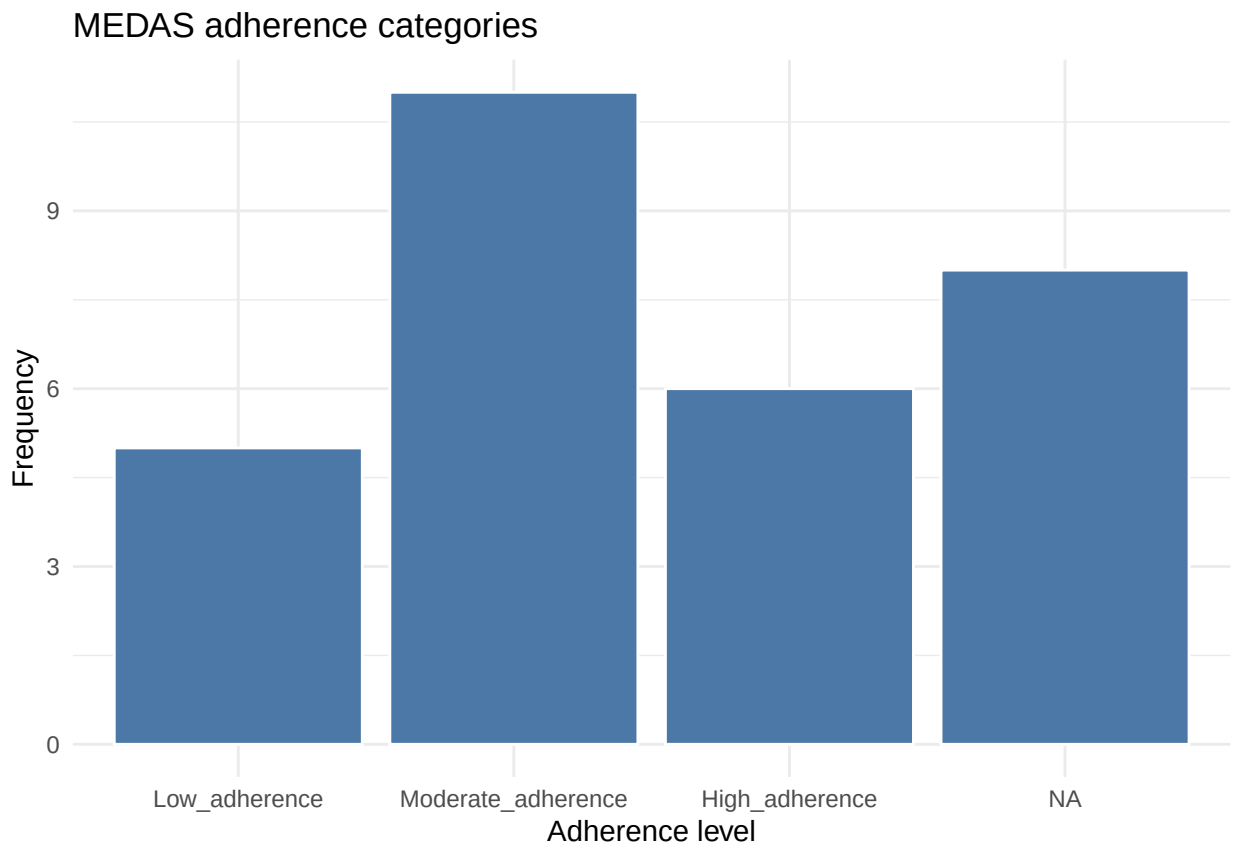
```r
ggplot(edss_long, aes(x = time, y = EDSS_val, color = physical_activity, group = physical_activity)) +
  stat_summary(fun = median, geom = "line", linewidth = 1.2, na.rm = TRUE) +
  stat_summary(fun = median, geom = "point", size = 3, na.rm = TRUE) +
  theme_minimal() + labs(title="EDSS during time by Physical Activity", x = "Time", y = "Median EDSS")
```



4. Summary graph of the distribution of the MEDAS variable in the cohort:

```r
merged_data$MEDAS_class <- factor(case_when(
  merged_data$MEDAS < 40 ~ "Low_adherence",
  merged_data$MEDAS >= 40 & merged_data$MEDAS < 60 ~ "Moderate_adherence",
  merged_data$MEDAS >= 60 ~ "High_adherence",
  TRUE ~ NA_character_),
  levels = c("Low_adherence", "Moderate_adherence", "High_adherence"),ordered = TRUE)

ggplot(merged_data, aes(x = MEDAS_class)) + geom_bar(fill = "#4C78A8", color = "white") + theme_minimal
```



4.1. EDSS and MEDAS, non-significant difference between groups at all time points.

```
# EDSS vs MEDAS
kruskal.test(as.numeric(as.character(EDSS_T0)) ~ MEDAS, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T0)) by MEDAS
## Kruskal-Wallis chi-squared = 6.0573, df = 7, p-value = 0.5331
```

```
kruskal.test(as.numeric(as.character(EDSS_T12)) ~ MEDAS, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T12)) by MEDAS
## Kruskal-Wallis chi-squared = 6.3061, df = 8, p-value = 0.613
```

```
kruskal.test(as.numeric(as.character(EDSS_T24)) ~ MEDAS, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T24)) by MEDAS
## Kruskal-Wallis chi-squared = 4.3342, df = 6, p-value = 0.6315
```

```
kruskal.test(as.numeric(as.character(EDSS_T36)) ~ MEDAS, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T36)) by MEDAS
## Kruskal-Wallis chi-squared = 3, df = 3, p-value = 0.3916
```

Exploratory analysis of some relevant clinical-radiological variables 1. Age at T0 and relapse, non-significant difference between groups at all time points.

```
# age_at_t0 vs relapse
t.test(age_at_t0 ~ relapse_T6, data = merged_data)
```

```
##
##  Welch Two Sample t-test
##
## data:  age_at_t0 by relapse_T6
## t = -0.10904, df = 5.3868, p-value = 0.9171
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -10.890771   9.986009
## sample estimates:
##  mean in group No mean in group Yes
##          35.21429          35.66667
```

```
t.test(age_at_t0 ~ relapse_T24, data = merged_data)
```

```
##
##  Welch Two Sample t-test
##
## data:  age_at_t0 by relapse_T24
## t = -0.48521, df = 1.1154, p-value = 0.7056
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -134.5081  122.0415
## sample estimates:
##  mean in group No mean in group Yes
##          36.26667          42.50000
```

2. Age at T0 and lesions, positive and significant difference between groups on T6 an T12.

```
# age_at_t0 vs lesions
t.test(age_at_t0 ~ lesioni_T6, data = merged_data) # correlazione SIGNIFICATIVA (di poco)
```

```
##
##  Welch Two Sample t-test
##
## data:  age_at_t0 by lesioni_T6
## t = 2.1402, df = 14.815, p-value = 0.0494
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##   0.01636283 10.85030384
## sample estimates:
##  mean in group No mean in group Yes
##          35.93333          30.50000
```

```
t.test(age_at_t0 ~ lesioni_T12, data = merged_data) # correlazione SIGNIFICATIVA
```

```
##
```

```
##  Welch Two Sample t-test
##
## data:  age_at_t0 by lesioni_T12
## t = -3.616, df = 4.8659, p-value = 0.01602
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -20.356548  -3.357738
## sample estimates:
##  mean in group No mean in group Yes
##          34.47619          46.33333
```

```
t.test(age_at_t0 ~ lesioni_T24, data = merged_data) # correlazione non significativa
```

```
##
##  Welch Two Sample t-test
##
## data:  age_at_t0 by lesioni_T24
## t = -1.2185, df = 2.6242, p-value = 0.3212
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -38.82454  18.58644
## sample estimates:
##  mean in group No mean in group Yes
##          35.21429          45.33333
```

3. EDSS and lesion burden, non-significant difference between groups at all time points, but looking at the graph you can notice a possible trend of association of higher EDSS values to patients with lesions.

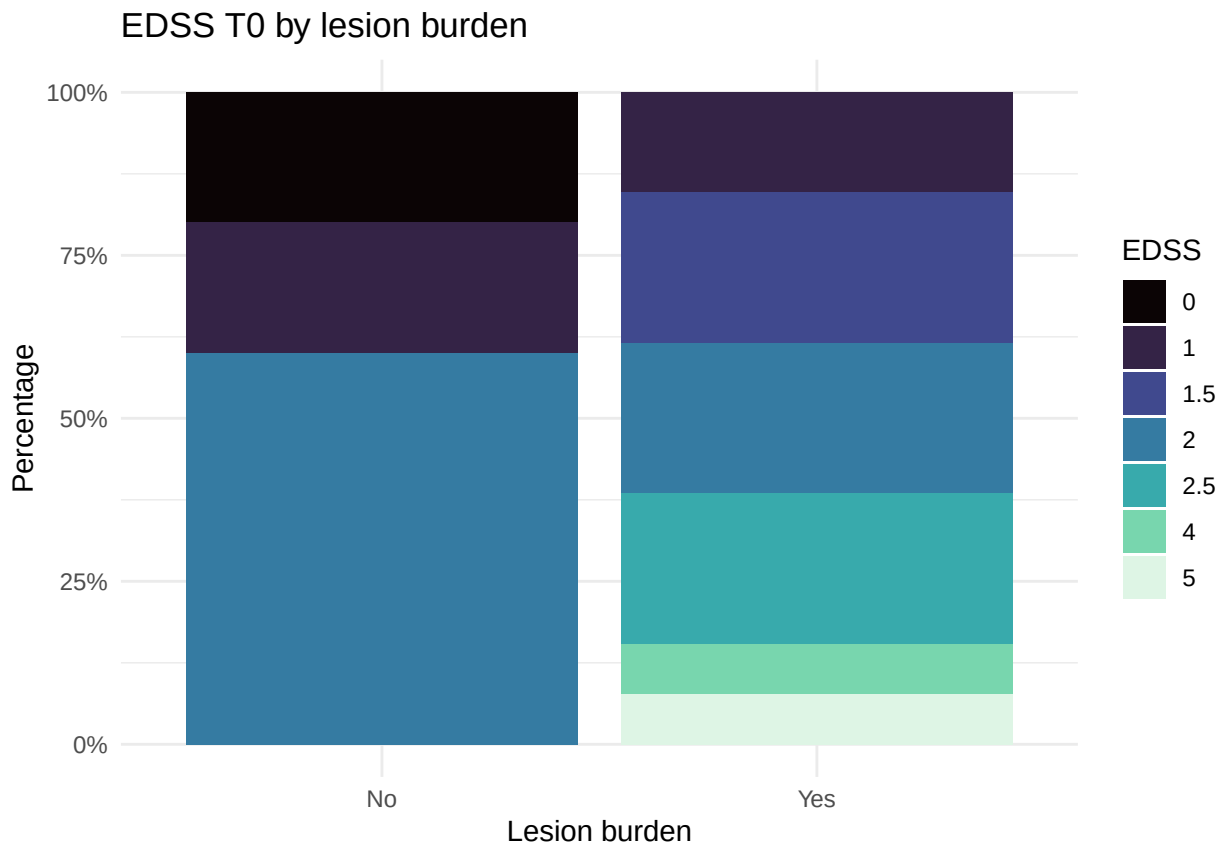
```
#lesion burden
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ lesion_burden, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by lesion_burden
## W = 20.5, p-value = 0.2452
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ lesion_burden, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by lesion_burden
## W = 21.5, p-value = 0.3911
## alternative hypothesis: true location shift is not equal to 0
```

```
merged_data %>%
  filter(!is.na(EDSS_T0), !is.na(lesion_burden)) %>%
  mutate(
    EDSS_T0 = as.factor(EDSS_T0),
    lesion_burden = factor(lesion_burden, labels = c("No", "Yes"))) %>%
  ggplot(aes(x = lesion_burden, fill = EDSS_T0)) +
  geom_bar(position = "fill") + scale_y_continuous(labels = scales::percent) +
  scale_fill_viridis_d(option = "mako") + theme_minimal() + labs(title = "EDSS T0 by lesion burden", x =
```



4. EDSS and bone marrow lesions, non-significant difference between groups at all time points, but looking at the graph you can notice a trend of association of higher EDSS values to patients with lesions.

```
#Bone marrow lesions
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ bone_marrow_lesions, data = merged_data, exact = FALSE)
```

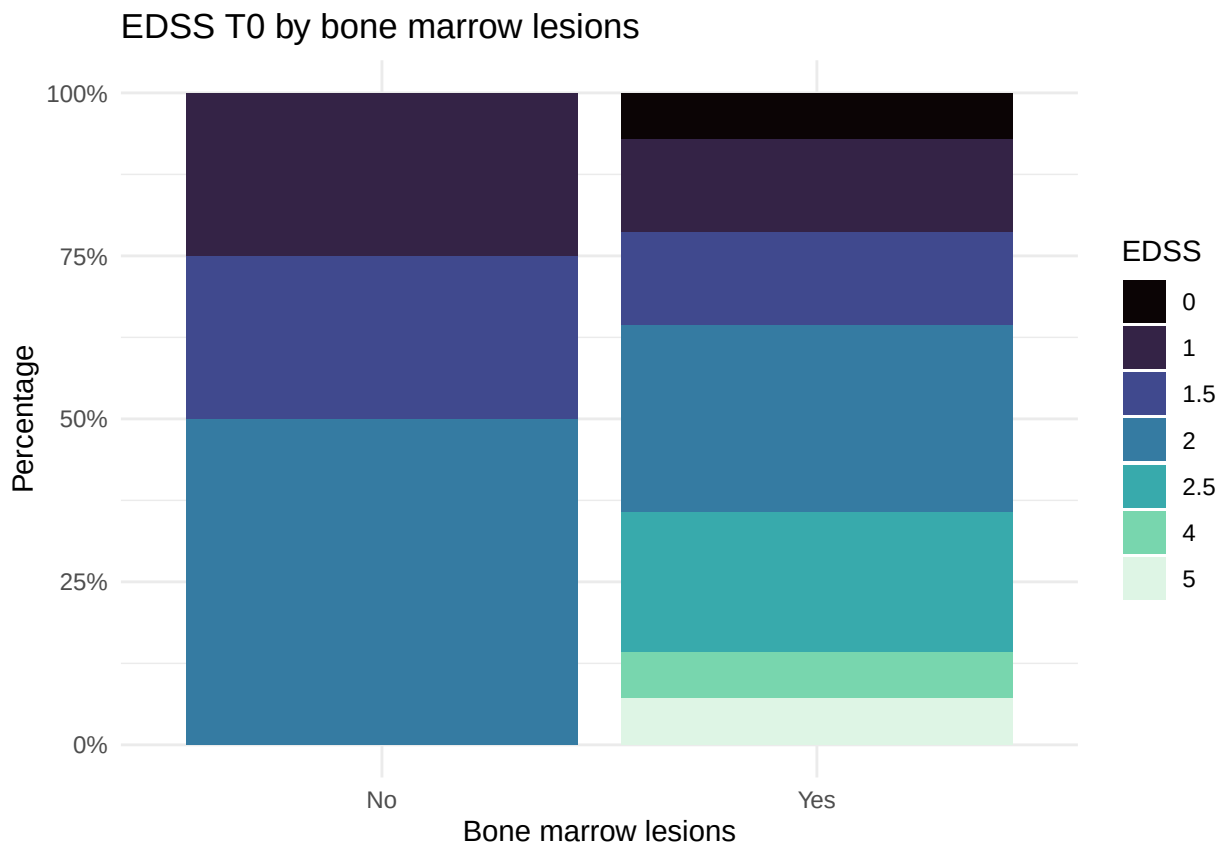
```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by bone_marrow_lesions
## W = 20, p-value = 0.4142
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ bone_marrow_lesions, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by bone_marrow_lesions
## W = 6, p-value = 0.02474
## alternative hypothesis: true location shift is not equal to 0
```

```
merged_data %>%
  filter(!is.na(EDSS_T0), !is.na(bone_marrow_lesions)) %>%
  mutate(
    EDSS_T0 = as.factor(EDSS_T0),
    bone_marrow_lesions = factor(bone_marrow_lesions, labels = c("No", "Yes"))) %>%
  ggplot(aes(x = bone_marrow_lesions, fill = EDSS_T0)) +
```

```
  geom_bar(position = "fill") + scale_y_continuous(labels = scales::percent) +
  scale_fill_viridis_d(option = "mako") + theme_minimal() + labs(title = "EDSS T0 by bone marrow lesions
```



5. EDSS and subtentorial lesions, non-significant difference between groups at all time points.

```
#subtentorial lesions
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ subtentorial_lesions, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by subtentorial_lesions
## W = 32, p-value = 0.9591
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ subtentorial_lesions, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by subtentorial_lesions
## W = 24, p-value = 0.7289
## alternative hypothesis: true location shift is not equal to 0
```

6. EDSS and gadolinium contrast, non-significant difference between groups at all time points.

```
#gadolinium contrast
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ gadolinium_contrast, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by gadolinium_contrast
## W = 31.5, p-value = 0.744
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ gadolinium_contrast, data = merged_data, exact = FALSE)
```

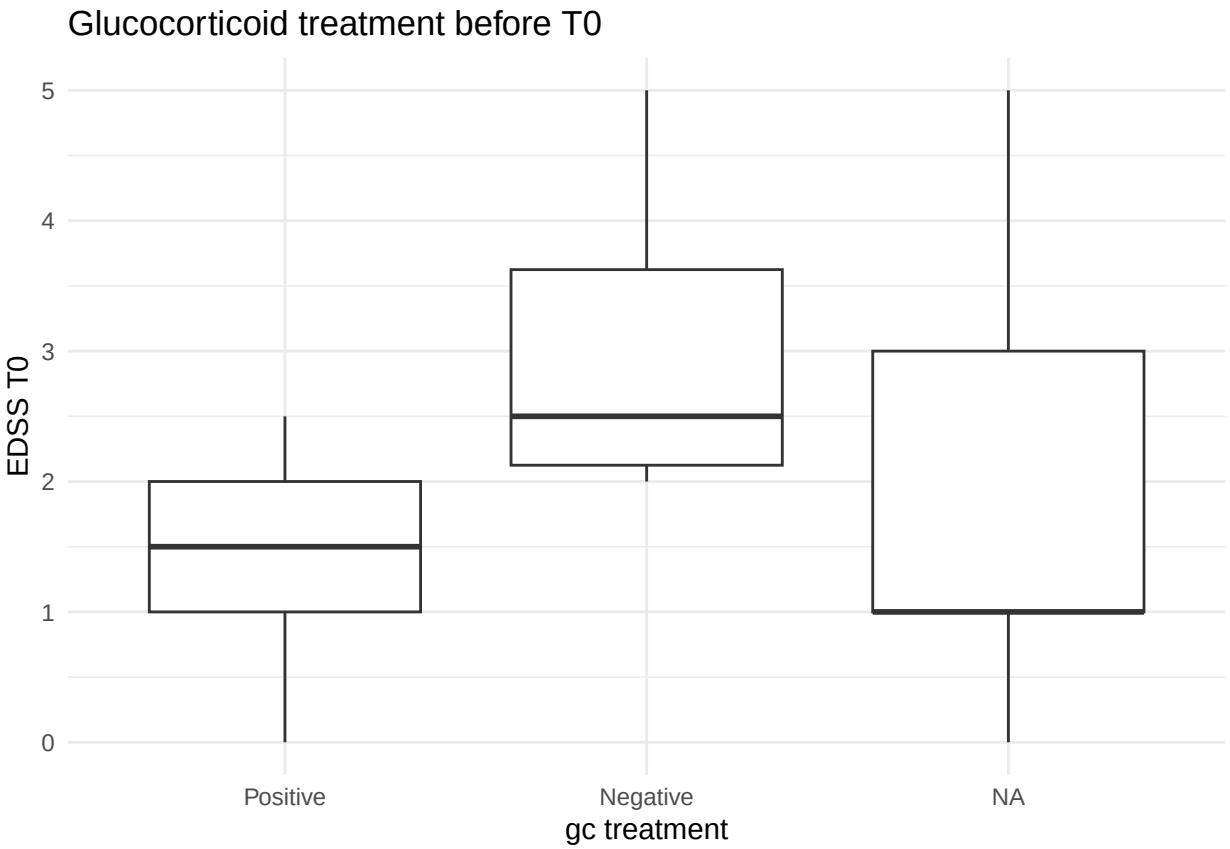
```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by gadolinium_contrast
## W = 25, p-value = 0.6538
## alternative hypothesis: true location shift is not equal to 0
```

7. Glucocorticoid treatment before baseline and EDSS, significant positive difference between groups in the T0 timepoint.

```
#glucocorticoid treatment before T0 e edss T0 #SIGNIFICATIVO
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ gc_treatment, data = merged_data)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by gc_treatment
## W = 7, p-value = 0.00621
## alternative hypothesis: true location shift is not equal to 0
```

```
ggplot(merged_data, aes(x = gc_treatment, y = as.numeric(as.character(EDSS_T0)))) + geom_boxplot() + th
```



Overall, no statistically significant associations were observed between EDSS and most clinical or lifestyle variables across time points. Physical activity showed a potential trend, with more sedentary patients displaying higher EDSS values over time. Age at baseline was positively associated with EDSS only at later time points, although these results may be influenced by missing data.